

Section Report 4

1. The tool falco was first used to analyze the sequence quality at a faster rate.
2. The fastp tool was used on the transcriptome reads to do a quality read and filtering on poor quality reads. Once it is run in Galaxy a summary will show the number of base pairs before and after filtering, so how many basepairs were kept and how many were removed due to poor quality. It will also show adapter dimers, which are sequences that are not good and can cause errors when processing transcriptome and should therefore be removed. For this transcriptome, the filtering result showed that there were zero reads with adapter dimers, and should be essentially good to continue with the next step.
3. The purpose of doing HISAT2 is to align your genome with the reference read and mapping the sequence out by identifying the points that align better.
4. The purpose of StringTie is to take the aligned sequences and assemble them into transcripts that will contain specific and essentially better genes. In other words, StringTie will assemble and accurately complete genes that will better show their gene expression or function of the genes.
5. The purpose of StringTie Merge is to assemble the multiple transcripts from StringTie into a transcriptome for differential analysis.
6. The tool featureCount is used to measure the gene expression in the alignment file from HISAT which should correspond to the history reference genome used for the alignment.
7. The purpose of using DESeq2 is to determine the different genomic expressions from the count tables generated by feature counts. The normalized count in DESeq2 allows for genes to be compared across the sample which are the wild types and mutants.
8. DESeq2 Plots
 - a. The PCA plot analysis shows the mutant samples clustered together on the right and the wild type samples on the left side except that wildtype 2 is farther away from the other two wildtypes which means that wildtype2 may have more variation than the other two. Overall, both groups show different gene expressions, which can be supported by the 94% variance between the mutants and wildtypes. The 3% shows the variation between replicates which is a small percent that suggests that the closer they are similar. For example, the mutants are clustered really close together and therefore shows that they are good replicates that are clustered properly.
 - b. The sample to sample distance plot also shows each group's gene expression similarities and differences. Both wildtype and mutant replicates are clustered separately which are shown by the dark blue, and show replicates are very

similar with one another. The lighter colors on the graph show that samples are different and so, gene expression between the two sample types vary greatly.

- c. The dispersion estimates graph shows most of the points following the fitted red curve. At a high dispersion, there is more variability and lower gene expression. In the graph, most dots are at a lower dispersion, indicating strong gene expression.
 - d. The histogram of the p-values for wild type and mutant show that genes have a low p value that are statistically significant indicated by the higher bar in the graph and support that there is differential expression between the mutant and wildtype samples.
 - e. In the MA- plot, the blue dots are the ones that are significant because they show differential gene expression, where the dots above 0 are upregulated and the ones below are downregulated which are less expressed than the upregulated ones.
9. The purpose of filtering the Deseq2 result file is to select for the p-values that are less than 0.05 which are considered significant and the null hypothesis is rejected. In addition, the $\log_2(\text{FC})$ has values that are negative and so the absolute value has to be taken and be greater than 1 to find the genes that have significant expressions or are the ones that are biologically important to further look at and interpret their function.
 10. The purpose of concatenating multiple datasets or collections is to join the uploaded header with the most recent filter dataset which have the p-value and $\log_2(\text{FC})$ adjusted.
 11. The volcano plot shows what genes are expressed differently by looking at the upregulated genes (red dots) which are expressed more and the downregulated genes (blue dots) which are expressed less. In the graph, the higher the dots, like the red ones higher up, show that they are statistically more significant than the lower red dots in the graph. In addition, the graph shows that genes that are farther apart from the center show more changes in gene expression.
 12. The heatmap2 shows the gene expression in mutant and wildtype samples. The dark red and blue shows that the gene is highly expressed, while the lighter colors indicate a lower gene expression. Some genes are highly expressed across both the mutants and the wildtypes and some only expressed on the wildtypes. For example, gene-BTH_RS35340 was highly expressed throughout all samples of both mutants and wild types. However, there were also other genes in just wildtypes that were highly expressed only. In addition, there were some genes that were not even expressed.
 13. Finally, the last steps (18-23) using filtering and cutting columns is to organize the data for easier interpretation of the up and down regulated genes with their function.
 14. Explanation of one up and down regulated gene and how it contributes to differential phenotypes.

- a. One upregulated gene is gene-BTH_RS25845 with a function of producing an ATP binding protein. This protein may be important since it needs the gene to produce a protein that needs ATP in order for it to do or complete its purpose in a pathway. Looking at the heatmap, this specific gene is only seen in the wildtype samples and highly expressed in the wildtypes. For these genes, there were no expressions of it in mutants and show that they have essentially a decrease in this gene function or there may be a mutation in which this gene cannot be expressed.
- b. One downregulated gene is gene-BTH_RS08210 that produces an enzyme, UxaA family hydrolase, which is used in a specific pathway in which this enzyme is used for energy production. This gene is expressed throughout both samples, the mutant and the wildtypes. This means that the gene is important because it is highly expressed in both the samples.